

SRS – Software Requirements Specification

1. Document Information

Field	Details
Project Name	Cravio – Online Food Delivery Platform
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2. System Context

Cravio is an online food delivery software platform that connects customers, restaurant partners, and delivery partners through a centralized system.

The system will support the complete order lifecycle, starting from restaurant and menu discovery through order placement, payment, restaurant processing, delivery assignment, order tracking, and final delivery.

The platform will interact with external services such as payment gateways, notification services, and location or mapping services where required.

The system will provide role-based access so that each user can access only the functions relevant to their role.

3. Detailed Functional Specifications

3.1 User Registration & Login

Requirement ID	Detailed Specification
SRS-FR-01	The system shall provide a registration interface where users can enter their mobile number or email address.
SRS-FR-02	The system shall validate whether the entered mobile number or email address is in a valid format.
SRS-FR-03	The system shall verify that the mobile number or email address is not already registered before creating a new account.
SRS-FR-04	The system shall create a user account only after successful validation of the required registration information.
SRS-FR-05	The system shall provide a login interface for registered users to enter their credentials.
SRS-FR-06	The system shall validate the entered credentials against the registered user account.
SRS-FR-07	The system shall display an appropriate error message when invalid credentials are entered.
SRS-FR-08	The system shall allow successfully authenticated users to access functions permitted by their assigned role.
SRS-FR-09	The system shall prevent unauthenticated users from accessing protected functions such as checkout and order placement.

3.2 Restaurant & Menu

Requirement ID	Detailed Specification
SRS-FR-10	The system shall retrieve restaurant information based on the customer's selected location and current restaurant availability.

SRS-FR-11	The system shall maintain restaurant data including restaurant ID, name, address, contact information, rating, operating status, and service availability.
SRS-FR-12	The system shall retrieve menu data associated with the selected restaurant.
SRS-FR-13	Each menu item shall contain an item ID, item name, description, price, category, and availability status.
SRS-FR-14	The system shall display only menu items that are currently available for ordering.
SRS-FR-15	When a restaurant user updates an item's price or availability, the system shall store the updated information and use the latest valid data for subsequent customer orders.
SRS-FR-16	The system shall validate that a selected menu item is still available before adding it to the customer's cart.
SRS-FR-17	If a selected item becomes unavailable before it is added to the cart, the system shall prevent the addition and display an appropriate message to the customer.

3.3 Search & Discovery

Requirement ID	Detailed Specification
SRS-FR-18	The system shall accept a customer's search input for restaurant names, food item names, or relevant keywords.
SRS-FR-19	The system shall process the search input and return matching restaurants or food items available to the customer.
SRS-FR-20	The system shall support filtering of search results using available criteria such as restaurant availability, food category, rating, or price range.

SRS-FR-21	The system shall apply the selected filters together with the customer's search criteria when generating results.
SRS-FR-22	The system shall display the number of matching results or an appropriate indication when results are available.
SRS-FR-23	When no matching results are found, the system shall display an appropriate message and allow the customer to modify the search criteria.
SRS-FR-24	The system shall remove or reset an applied filter when the customer chooses to clear the filter.

3.4 Cart Management

Requirement ID	Detailed Specification
SRS-FR-25	The system shall create or update a customer's active cart when an available food item is added.
SRS-FR-26	The system shall store the selected food item, quantity, unit price, and applicable item-level details in the cart.
SRS-FR-27	The system shall calculate the item subtotal using the applicable unit price and selected quantity.
SRS-FR-28	The system shall calculate the cart total based on the current items and their applicable prices.
SRS-FR-29	When a customer changes an item's quantity, the system shall validate the new quantity before updating the cart.
SRS-FR-30	The system shall recalculate the affected item subtotal and overall cart total after a valid quantity change.
SRS-FR-31	When a customer removes an item, the system shall remove it from the active cart and recalculate the cart total.
SRS-FR-32	The system shall prevent checkout when the active cart contains no valid orderable items.

SRS-FR-33	Before checkout, the system shall revalidate item availability and applicable prices to ensure that the cart contains current order information.
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3.5 Order Placement

Requirement ID	Detailed Specification
SRS-FR-34	The system shall allow checkout only for an authenticated customer with a valid, non-empty cart.
SRS-FR-35	During checkout, the system shall retrieve and display the current cart items, quantities, applicable prices, delivery address, and order amount.
SRS-FR-36	The system shall validate that all mandatory delivery information is provided before order submission.
SRS-FR-37	The system shall revalidate cart items, availability, and applicable prices before creating the order.
SRS-FR-38	The system shall calculate the final order amount using the applicable item amounts and configured delivery charges or other applicable charges.
SRS-FR-39	The system shall create a unique order record when the order passes all required validations and the applicable payment condition is satisfied.
SRS-FR-40	The system shall store the order ID, customer details, restaurant details, ordered items, quantities, order amount, delivery details, payment status, and order status.
SRS-FR-41	The system shall assign the initial order status as Order Placed after successful order creation.
SRS-FR-42	The system shall display the generated order ID and order confirmation details to the customer after successful order placement.

3.6 Payment Processing

Requirement ID	Detailed Specification
SRS-FR-43	The system shall display the payment methods available for the selected order based on applicable business rules.
SRS-FR-44	For online payment, the system shall send the required payment information to the configured payment gateway.
SRS-FR-45	The system shall generate or maintain a unique payment reference associated with the order.
SRS-FR-46	The system shall process the payment gateway response and record the resulting payment status.
SRS-FR-47	The system shall mark an online payment as successful only after receiving a valid successful payment confirmation from the payment service.
SRS-FR-48	If an online payment fails, the system shall retain the order/payment information required for retry or appropriate failure handling without creating a duplicate order.
SRS-FR-49	For Cash on Delivery, the system shall record the payment method as COD and maintain the applicable pending payment status until delivery completion or business-defined settlement.
SRS-FR-50	The system shall associate the payment status and payment reference with the corresponding order record.
SRS-FR-51	The system shall prevent unauthorized users from accessing or modifying payment information.

3.7 Order Tracking & Status Management

Requirement ID	Detailed Specification
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SRS-FR-52	The system shall maintain the current status of each order throughout its lifecycle.
SRS-FR-53	The system shall support the following status sequence: Order Placed → Confirmed → Preparing → Picked Up → Out for Delivery → Delivered.
SRS-FR-54	The system shall validate each requested status transition against the permitted order workflow before updating the order.
SRS-FR-55	The system shall record the date and time whenever an order status is changed.
SRS-FR-56	The system shall maintain the previous status information as part of the order status history.
SRS-FR-57	The system shall display the current order status and relevant status history to the customer.
SRS-FR-58	The system shall allow only authorized users or system processes to perform status updates applicable to their role.
SRS-FR-59	The system shall prevent an order from being marked as Delivered unless the required delivery completion conditions are satisfied.

3.8 Restaurant Order Management

Requirement ID	Detailed Specification
SRS-FR-60	The system shall make newly created orders available to the restaurant associated with the order.
SRS-FR-61	The system shall display relevant order information to authorized restaurant users, including order ID, ordered items, quantities, customer delivery details, and order time.
SRS-FR-62	The system shall allow an authorized restaurant user to accept or reject an order.

SRS-FR-63	When a restaurant accepts an order, the system shall update the order status to Confirmed .
SRS-FR-64	When a restaurant rejects an order, the system shall record the rejection and prevent the order from proceeding through the normal preparation workflow.
SRS-FR-65	The system shall allow authorized restaurant users to update the order to the Preparing status after acceptance.
SRS-FR-66	The system shall record the relevant date and time for restaurant order acceptance, rejection, and preparation status updates.
SRS-FR-67	The system shall prevent restaurant users from modifying orders that do not belong to their restaurant.

3.9 Delivery Management

Requirement ID	Detailed Specification
SRS-FR-68	The system shall identify eligible delivery partners for an accepted order based on configured assignment rules.
SRS-FR-69	The system shall create a delivery assignment associated with the relevant order and selected delivery partner.
SRS-FR-70	The system shall provide the assigned delivery partner with the required order and pickup information.
SRS-FR-71	The system shall allow the assigned delivery partner to accept or acknowledge the delivery assignment.
SRS-FR-72	The system shall update the order status to Picked Up when the delivery partner confirms that the order has been collected from the restaurant.
SRS-FR-73	The system shall allow the assigned delivery partner to update the delivery status to Out for Delivery after

	pickup.
SRS-FR-74	The system shall record the relevant date and time for delivery assignment, acknowledgement, pickup, and delivery status updates.
SRS-FR-75	The system shall display applicable delivery status updates to the customer.
SRS-FR-76	The system shall prevent a delivery partner from accessing or updating deliveries that are not assigned to them.

3.10 Notification Management

Requirement ID	Detailed Specification
SRS-FR-77	The system shall create a notification event when a configured order-related event occurs.
SRS-FR-78	The system shall determine the appropriate notification recipient based on the event and user role.
SRS-FR-79	The system shall send an order confirmation notification to the customer after successful order placement.
SRS-FR-80	The system shall send a notification to the restaurant when a new order is successfully created.
SRS-FR-81	The system shall notify the customer when the order status changes to a configured status.
SRS-FR-82	The system shall notify the assigned delivery partner when a delivery assignment is created.
SRS-FR-83	The system shall notify the customer when an order is rejected or cancelled.
SRS-FR-84	The system shall use the notification channel configured for the relevant event, such as in-app notification, SMS, or email.

SRS-FR-85	The system shall record the notification status to support monitoring of successfully sent or failed notifications.
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3.11 Cancellation & Refund Management

Requirement ID	Detailed Specification
SRS-FR-86	The system shall allow a customer to initiate an order cancellation request when the order is in a cancellable state.
SRS-FR-87	The system shall validate the customer's cancellation request against the applicable cancellation rules before processing it.
SRS-FR-88	If cancellation is permitted, the system shall update the order status to Cancelled and record the cancellation date, time, and applicable reason.
SRS-FR-89	If cancellation is not permitted, the system shall reject the request and display an appropriate message to the customer.
SRS-FR-90	For an eligible prepaid order, the system shall create a refund request according to the configured refund policy.
SRS-FR-91	The system shall maintain the refund status associated with the cancelled prepaid order.
SRS-FR-92	The system shall prevent duplicate refund requests for the same cancelled payment.
SRS-FR-93	The system shall notify the customer about the cancellation outcome and applicable refund status.

3.12 Reporting & Access Management

Requirement ID	Detailed Specification
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SRS-FR-94	The system shall provide authorized users with access to configured business reports.
SRS-FR-95	The system shall retrieve report data from the relevant order, payment, restaurant, and delivery records.
SRS-FR-96	The system shall provide order-related information such as order count, order status, and order details in applicable reports.
SRS-FR-97	The system shall provide revenue and payment-related information to users authorized to view financial reports.
SRS-FR-98	The system shall provide restaurant and delivery performance information in the applicable reports.
SRS-FR-99	The system shall restrict report access according to the authenticated user's role and assigned permissions.
SRS-FR-100	The system shall prevent unauthorized users from viewing restricted business or financial information.
SRS-FR-101	The system shall display an appropriate message when a report cannot be generated because the requested data is unavailable or an error occurs.

3.13 — Data Requirements

Data Entity	Key Data Fields	Purpose
User	User ID, Name, Mobile/Email, Role, Account Status	User account and access management
Restaurant	Restaurant ID, Name, Address, Contact, Rating, Availability Status	Restaurant information

Menu Item	Item ID, Restaurant ID, Name, Description, Category, Price, Availability	Food and menu management
Cart	Cart ID, User ID, Item ID, Quantity, Unit Price, Cart Total	Temporary order selection
Order	Order ID, User ID, Restaurant ID, Items, Amount, Delivery Address, Order Status, Order Date/Time	Order processing and tracking
Payment	Payment ID, Order ID, Payment Method, Payment Reference, Amount, Payment Status	Payment transaction tracking
Delivery	Delivery ID, Order ID, Delivery Partner ID, Assignment Status, Delivery Status, Timestamps	Delivery management
Notification	Notification ID, User ID, Event Type, Channel, Status, Created Date/Time	Notification tracking

3.14 External Interfaces & Integrations

Interface / Service	Integration Requirement	Purpose
Payment Gateway	Cravio shall communicate with the configured payment	Online payment processing

	gateway to initiate online payments and receive payment status responses.	
Notification Service	Cravio shall integrate with configured notification services for sending SMS, email, push, or in-app notifications where applicable.	Customer, restaurant, and delivery notifications
Location / Mapping Service	Cravio may integrate with a mapping or location service to support address validation, delivery-area checks, and location-based delivery operations.	Location and delivery support
Authentication Service	Cravio may use an authentication mechanism or service to securely verify user credentials and maintain authenticated sessions.	User authentication and access control

4. User Roles & Access Control

Role	Access & Permissions
Customer	The customer shall be able to manage their account, browse restaurants and menus, manage the cart, place orders, make payments, track orders, and request eligible cancellations.

Restaurant Partner	The restaurant partner shall be able to manage permitted restaurant and menu information, view received orders, accept or reject orders, and update preparation-related status.
Delivery Partner	The delivery partner shall be able to view assigned deliveries, acknowledge assignments, and update applicable delivery statuses.
Authorized Business/Admin User	The authorized user shall be able to access permitted business reports, manage applicable system information, and administer role-based access according to assigned permissions.

Access Control Rules

- The system shall authenticate users before allowing access to protected functions.
- The system shall authorize each requested action based on the user's assigned role.
- The system shall prevent users from accessing functions or data outside their assigned permissions.
- The system shall maintain access restrictions for sensitive information such as payment and business reports.

5. Detailed Business Rules & Validations

Rule ID	Detailed Rule / Validation
SRS-BR-01	The system shall verify that the customer is authenticated before allowing checkout or order placement.
SRS-BR-02	The system shall validate item availability before adding an item to the cart and again before order creation.
SRS-BR-03	The system shall prevent an order from moving to Preparing unless the restaurant has accepted the order.
SRS-BR-04	The system shall allow restaurant order actions only to authorized users associated with the respective restaurant.

SRS-BR-05	The system shall allow delivery status updates only from the delivery partner assigned to the relevant order.
SRS-BR-06	The system shall validate order status and applicable cancellation conditions before processing a cancellation request.
SRS-BR-07	The system shall initiate refunds only for eligible prepaid orders and shall prevent duplicate refund processing.
SRS-BR-08	The system shall restrict business and financial reports to users with the required permissions.
SRS-BR-09	The system shall validate permitted order-status transitions and prevent invalid status changes.
SRS-BR-10	The system shall enforce role-based access for protected functions and data throughout the platform.

6. Error & Exception Handling

Error ID	System Behaviour
SRS-EH-01	If invalid login credentials are entered, the system shall reject the login attempt and display an appropriate error message.
SRS-EH-02	If a selected food item becomes unavailable, the system shall prevent the item from being ordered and inform the customer.
SRS-EH-03	If an online payment fails or times out, the system shall display the payment failure status and shall prevent the same transaction from being recorded as successfully paid.
SRS-EH-04	If mandatory checkout information is missing or invalid, the system shall prevent order creation and identify the required information.
SRS-EH-05	If a restaurant rejects an order, the system shall update the order according to the applicable business rule and

	notify the customer.
SRS-EH-06	If a cancellation request does not meet the applicable conditions, the system shall reject the request and inform the customer.
SRS-EH-07	If a temporary system or external-service failure occurs during order processing, the system shall avoid creating duplicate orders or transactions.
SRS-EH-08	If a search returns no matching results, the system shall display an appropriate message and allow the customer to modify the search.
SRS-EH-09	If an unauthorized user attempts to access a restricted function, the system shall deny access and display an appropriate message.

7. Non-Functional Specifications

Requirement ID	Specification
SRS-NFR-01	The system shall provide responsive interfaces that function correctly across supported desktop, tablet, and mobile screen sizes.
SRS-NFR-02	Under normal operating conditions, commonly used operations such as restaurant search, menu loading, and cart updates shall complete within an agreed response-time target.
SRS-NFR-03	The system shall protect sensitive user and transaction data from unauthorized access.
SRS-NFR-04	The system shall use secure communication protocols for data transmitted between the client application and backend services.
SRS-NFR-05	The system shall maintain service availability in accordance with the agreed business availability target.

SRS-NFR-06	The system shall support horizontal or equivalent scaling as the number of customers, restaurants, orders, and delivery partners increases.
SRS-NFR-07	The system shall maintain consistency and accuracy of critical data across order, payment, restaurant, and delivery records.
SRS-NFR-08	The system shall maintain audit or application logs for important authentication, order, payment, and status-update activities.

8. Assumptions, Constraints & Dependencies

8.1 Assumptions

- Customers, restaurant partners, and delivery partners shall have access to a supported device and a stable internet connection.
- Restaurant partners shall maintain accurate menu, price, and item availability information.
- Customers shall provide valid contact and delivery information.
- Delivery partners shall operate within the service areas supported by Cravio.
- Required external services, such as payment and notification services, shall be available when needed.
- The initial release shall focus on the core food-ordering and delivery workflow.

8.2 Constraints

- Payment processing depends on the capabilities and availability of the selected payment gateway.
- Delivery availability may depend on location, service area, and delivery-partner availability.
- Notification delivery may depend on third-party notification services.
- Location-based functionality may depend on the selected mapping or location service.
- System functionality shall be limited by the permissions assigned to each user role.

8.3 Dependencies

- Payment Gateway
- Notification Service
- Location / Mapping Service
- Internet Connectivity

- Restaurant Availability and Data
- Delivery Partner Availability
- Stakeholder decisions and approval

9. Traceability & References

SRS Area	Source Document	Purpose
Functional Specifications	FRD – Functional Requirements Document	Detailed software behaviour based on approved functional requirements
User Roles & Access Control	FRD – User Roles & Permissions	Defines role-based system access
Business Rules & Validations	BRD + FRD	Converts business rules into system-level validation
Error & Exception Handling	FRD – Error & Exception Handling	Defines system response to expected errors and exceptions
Non-Functional Specifications	FRD – Non-Functional Requirements	Defines system quality and performance expectations
Data Requirements	Business & Functional Requirements	Defines major data entities required by the system
External Interfaces & Integrations	Business and Functional Requirements	Defines interaction with external services

9.2 Reference Documents

- Cravio – Project Overview
- Requirement Gathering – Initial Client Meeting
- MoM – Initial Client Requirement Meeting

- BRD - Business Requirements Document
- FRD - Functional Requirements Document